

MX-40 CNC Operation & Maintenance Manual

CNC Milling MX-40 5-Axis VMC — Rev 5.0

This document is a demonstration OEM-style manual generated for the SARVA-SENSE RAG troubleshooting assistant prototype.

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SECTION 5.4 - SPINDLE DIAGNOSTICS & ALARM CODES

ALARM CODE E101 (MX-40): Spindle Bearing Thermal Excursion / Temperature Warning. This code is specific to the MX-40 spindle cartridge and is NOT related to hydraulic pressure. Trigger: front spindle bearing thermocouple TC-SP1 reads above 78 degrees C for more than 60 seconds during continuous 5-axis cutting cycles.

Probable root causes of E101 on MX-40: (1) spindle coolant-through-tool flow restricted or coolant pump CP-3 underperforming, (2) front bearing grease depleted or contaminated with coolant, requiring bearing repack per the 2000-hour maintenance interval, (3) spindle running at sustained RPM above the duty-cycle rating for the current tool load, (4) thermocouple TC-SP1 sensor drift or wiring fault giving a false high reading.

Corrective Action Procedure for E101 (MX-40): Step 1 - reduce spindle RPM and pause the cutting cycle; allow the spindle to idle-cool for 5 minutes with coolant flow active. Step 2 - verify coolant-through-spindle flow rate at nozzle CT-1 meets the 4.5 L/min minimum specification; clean or replace the inline filter if flow is low. Step 3 - inspect and, if due, repack the front spindle bearing cartridge with OEM grease per the lubrication schedule. Step 4 - if TC-SP1 readings remain inconsistent with an infrared thermal check of the spindle nose, replace the thermocouple assembly TC-SP1.

ALARM CODE E205: Servo Drive Following Error on the X-axis ball screw. Corrective action: check ball screw preload and servo tuning parameters per Section 5.6.

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